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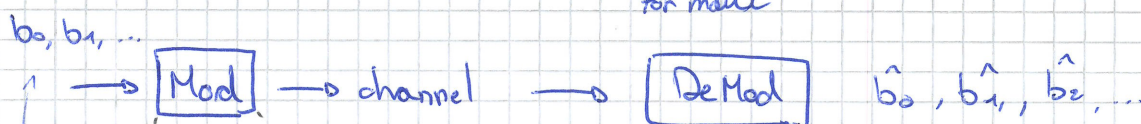
Wireless, Lecture 1 Discrete Time Model + Modulation

We have a bitstream: $b_0, b_1, \dots$

from A to B

discrete time $\rightarrow$ Continuous time $\rightarrow$ discrete time

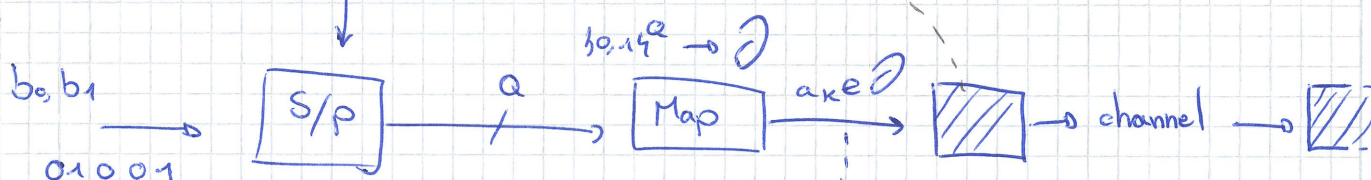
we do not care continuous for now



name of the bits that go in

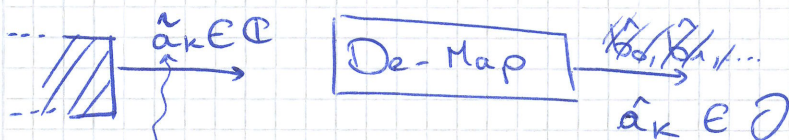
estimated bit, not necessarily equal to host-less bits

we want multiples bits in parallel



discrete time k

this would be in continuous time



this is not really an estimation, its just what I receive

① Mapper Maps Q -bits onto a Symbol

" Q -bits per channel use"

(more bits into one symbol $\uparrow \eta$)

$\{0, 1\}^Q \rightarrow \mathcal{D}$ = set of constellation points

$\{x_1, x_2, \dots, x_M\}$ w/ $M = 2^Q$ $x_n \in \mathbb{C}$

Map $\rightarrow$ bits to Symbols

Demap $\rightarrow$ number to Symbols to bits

$\rightarrow$ ② Channel $a_k \in \mathcal{D} \rightarrow \tilde{a}_k \in \mathbb{C}$

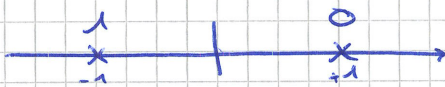
$\rightarrow$ ③ De mapper: ideally $\tilde{a}_k \in \mathcal{D}$ (we do not need additional step to recover info)

$\tilde{a}_k \in \mathbb{C}$ (in practice) due to noise

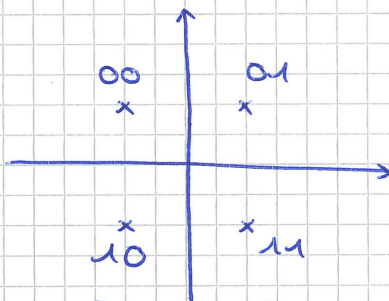
Constellation points we want a nbr that could represent

$Q = 1$

zero's and one's



1 bit per symbol



2 bits per symbol

QAM: $M = 2^Q$ $Q = 2, 4, 6, 8, \dots$

$Q = 4$ $\mathbb{R} \{x_i\} = \{\pm 1, \pm 3\}$ points should be evenly placed

if point are far from origin its good for precision but as $|A| \uparrow$ it requires more power

Signal Energy

$$P_{x_i} = |x_i|^2$$

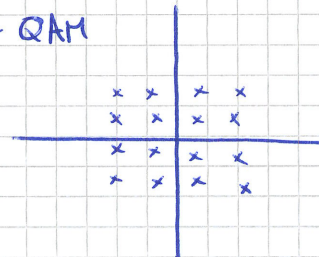
$$P_{a_k} = |a_k|^2$$

$\rightarrow$ this is per symbol energy

$\rightarrow$ we actually NEED the average Energy
all symbols equally likely

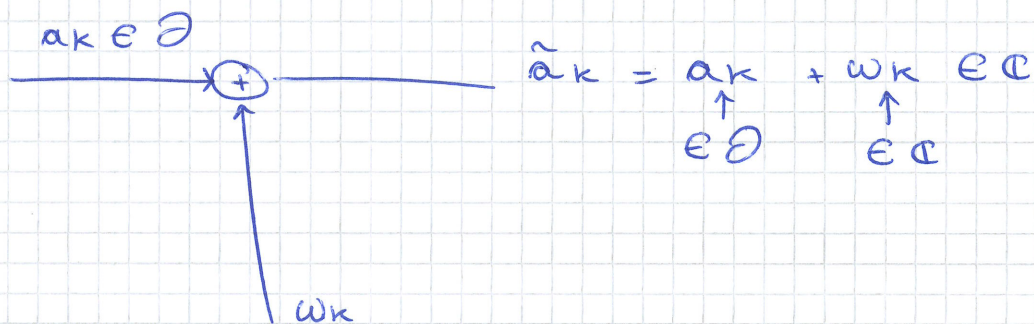
$$\bar{P} = \sum_{i=1}^M |x_i|^2 \cdot \underbrace{P_r(x_i)}_{\text{proba of } x_i} = \frac{1}{M} = \frac{2}{3} (M-1) \text{ just average energy}$$

16-QAM



②

The AWGN Additive White Gaussian Noise



we need 2 things

$$w_k \sim \mathcal{CN}(0, \sigma^2)$$

→ the variance of noise is the power

$$SNR = \frac{\bar{P}}{P_n} = \frac{\mathbb{E}\{|a_k|^2\}}{\mathbb{E}\{|w_k|^2\}} \leftarrow \bar{P}$$

we measure noise in [dB] $\leftarrow \sigma_w^2$

how do we compare things w/ Quality Metrics

$$SNR [dB] = 10 \log(\bar{P}/P_n)$$

Error-rate vs SNR

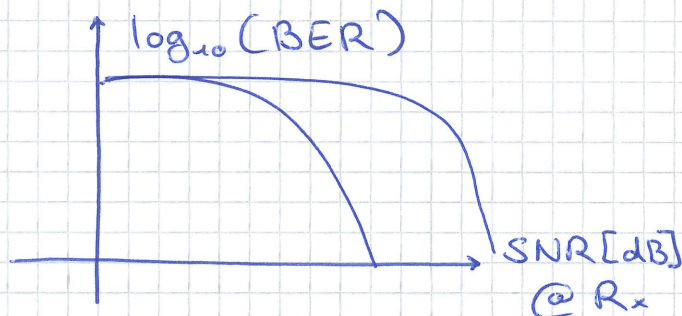
the Error Rate changes by orders of magnitude depending on SNR

linear axis = stupid

10^{-6} Wireless

10^{-15} Wireline

min values are incredibly small

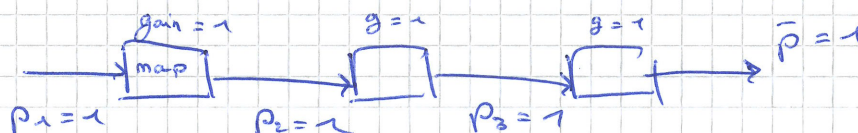


Ex: QPSK + 16 QAM (un-normalized)
Quadrature Phase Shift Keying → Quadrature Amplitude Modulation

$$\bar{P}(\text{QPSK}) < \bar{P}(16 \text{ QAM})$$

The error rate is not going to change, same distance between points

To avoid errors, in a block chain



the output power will always be 1 $\rightarrow \bar{p} \equiv 1$.

so this is why we used normalized constellation to keep avg power = 1.

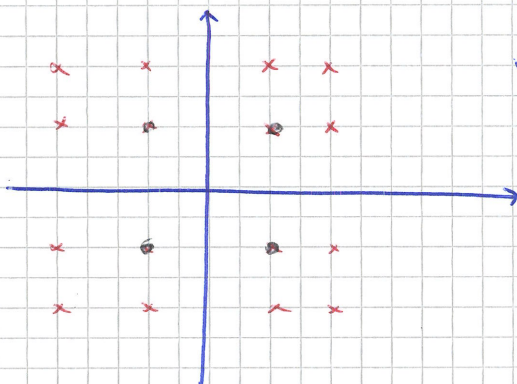
$$\bar{P}_{M-QAM} = \frac{2}{3} (M-1)$$

nice consequence: $SNR = \frac{1}{\sigma^2}$

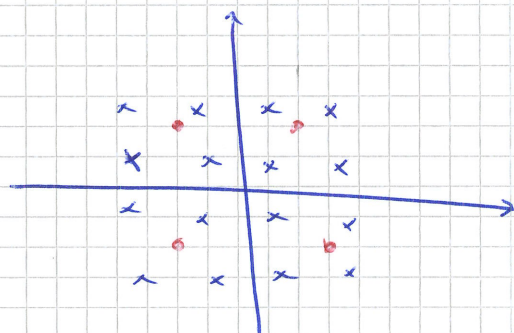
x_i = un-normalized constellation $\pm 1, \pm 3, \dots$

$$\bar{x}_i = \frac{1}{\sqrt{\bar{P}_{M-QAM}}} x_i$$

normalized constellation



normalized constellation $\rightarrow$



• QPSK

* 16-QAM

Noise

$$P(\omega) = \frac{1}{\sqrt{2\sigma^2\omega}} e^{-\frac{|\omega - m|^2}{2\sigma^2}}$$

* 16-QAM normalized

$\omega \in \mathbb{R}$

in practice $\sim \mathbb{C}$ noise

$$\omega = \omega_R + j \omega_I$$

$$\omega \sim \mathbb{C}N(0, \sigma^2)$$

$$E\{| \omega |^2\} = E\{| \omega_R |^2\} + E\{| \omega_I |^2\}$$

independent no cross term disappear

both are supposed to be the same

real & Complex noise are orthogonal and have same value amplitude

$$E\{| \omega |^2\} = 2 \times E\{| \omega_R |^2\}$$

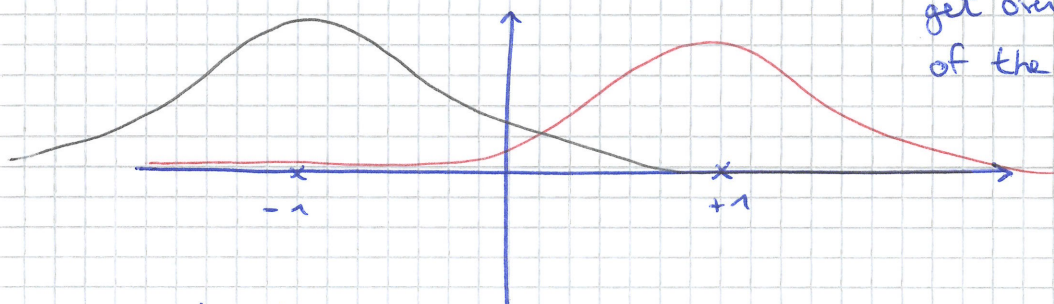
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$$w_R \sim N(0, \frac{\sigma^2}{2})$$

BPSK $Pr(a_k = \pm 1) = \frac{1}{2}$ same for w_I

$$Pr(\hat{a}_k) = Pr(a_k = 1) \cdot Pr(\tilde{a}_k | a_k = 1) + Pr(a_k = -1) \cdot Pr(\tilde{a}_k | a_k = -1)$$

Linear combination of two Gaussians to get small distribution of the noise



I want to minimize the probability of error, I want to make the choice that is most likely: Maximum Likelihood pred

ML - Principle

we chose some points that
max precision of prediction
no chose best constellation

$$\hat{a}_k = \arg \max_{x_i} \underbrace{Pr(a_k = x_i | \tilde{a}_k)}_{\text{we do not have this}}$$

BUT $Pr(\tilde{a}_k | a_k = x_i)$

we have this

$$Pr(\tilde{a}_k | a_k = x_i) \cdot Pr(a_k = x_i) = Pr(a_k | \tilde{a}_k) \cdot Pr(\tilde{a}_k)$$

↳ basically $\rightarrow Pr(a_k | \tilde{a}_k) = \frac{Pr(\tilde{a}_k | a_k) Pr(a_k)}{Pr(\tilde{a}_k)}$

$$Pr(\tilde{a}_k | a_k = x_i) = \frac{1}{2\sigma^2 M} e^{-\frac{|\tilde{a}_k - x_i|^2}{2\sigma^2}}$$

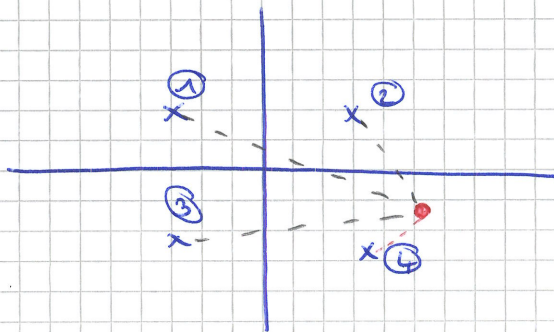
↳ the solution is to make
as small as possible

I do not need to find
value @ right we just
want to minimize influence
of noise

So Finally solution ML for AWGN

$$\arg \min_{x_i} \{ |\tilde{x}_k - x_i|^2 \}$$

Basically our estimation is: the closest point is the best estimation

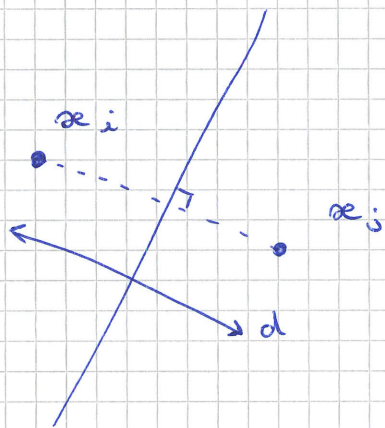


the best estimation of • is ④ as it is the closest point.

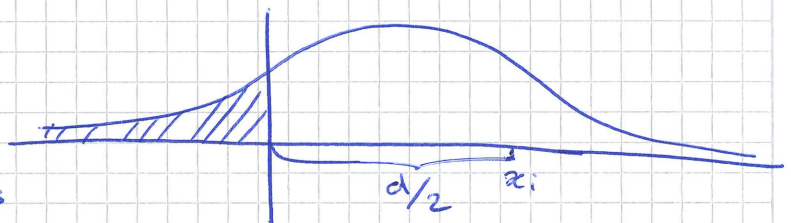
Error Probability

we have an error if $\tilde{x}_k$ lies closer to a point x_j if x_i was sent and $i \neq j$
(mistake one for another)

basically $|\tilde{x}_k - x_j|^2 < |\tilde{x}_k - x_i|^2$ easy to take $| \cdot |^2$ than $| \cdot |$
no sqrt problem



so we end up w/ 1D
Gaussian problem by projecting points in ortho dir



$$\Pr(w_k > \frac{d}{2}) = \int_{\frac{d}{2}}^{\infty} \text{Gauss}$$

$$= Q\left(\sqrt{\frac{d^2}{4} \frac{1}{\sigma^2}}\right)$$

Lo I can play with distance and power proportionally and have same error in the end

$$\text{SER (Simple Error Rate)} < Q\left(\sqrt{\frac{d_{\min}^2}{4} \frac{1}{\sigma^2}}\right) \times \frac{(M-1)}{\text{\# of error events}} \quad \text{③}$$

Symbol ~ Bounds on our error

SER > BER
no multiple bits/symbols